

limited to India. Globally, the significance of semiconductors wasn't fully appreciated until the last few years. Before the pandemic, the critical role of semiconductors in modern life was often overlooked. As states in India become more conscious of this, we can expect others who were slower to act to catch up quickly. The shift in awareness is timely, and the global and domestic momentum is moving in the right direction. This will help position India as a key player in the semiconductor industry, with a strong policy focus across multiple regions.

Do you think India's \$10 billion semiconductor scheme is sufficient to attract large-scale global investments when compared to the USA's \$52 billion CHIPS Act?

At this moment, we should not be comparing countries like A with B or C. Each nation needs to develop its own strategy and move at a pace that aligns with its unique circumstances and goals. With the industry growing from \$600 billion to a projected \$1 trillion in the next seven years, the demand for innovation and capacity is unprecedented. The key here is not just to spend as much as possible but to spend wisely, making smart investments that bring long-term results. Rather than focusing on competition between countries, the real challenge is ensuring that the world, as a whole, is prepared to meet this growing demand.

Following the path of the USA's restrictions and export ban, China, too, has decided to ban the export of germanium to other countries. How will this geopolitical scuffle affect the global growth of chips?

We must emphasise the need for diversification and redundancy in the semiconductor supply chain.

While the restrictions on critical minerals like germanium and indium will undoubtedly impact the industry, alternative sources for these materials could step up to fill the gaps. As the industry is expected to double in size over the next few years, it's clear that relying on a limited number of hubs is not sustainable. To manage the growing demand and mitigate risks—whether they stem from geopolitical tensions, climate issues, or other disruptions—it's essential to establish multiple semiconductor hubs across various regions. This would create the necessary redundancies so that the industry can shift production to other locations without severe disruptions if one country faces challenges. The interconnectedness of the global semiconductor industry is both a challenge and an opportunity, and the efforts to create a robust blueprint for the future are critical in ensuring its long-term sustainability.

Can you shed some insights into SEMI's upcoming strategies and the role of strategic partnerships in ensuring a sustainable and resilient supply chain?

We have to highlight the shift from the 'just-in-time' (JIT) philosophy to the 'just-in-case' (JIC) approach in the semiconductor supply chain. While JIT worked well for many years, offering cost-efficiency and lean inventories, the events of the last few years—especially the disruptions caused by the pandemic, geopolitical tensions, and climate challenges—have exposed the vulnerabilities in this model.

Companies and governments recognise that the price of preparedness is necessary to avoid the much higher cost of future supply chain breakdowns. In summary, the JIC approach, while more expensive

in the short term, is essential for ensuring long-term stability and preventing the kind of widespread disruptions that have impacted industries worldwide. As you mentioned, revisiting and redesigning the semiconductor supply chain is crucial for the industry's resilience.

By 2030, what is the total number of fabs, ATMPs, and OSAT units you are expecting in India?

While it may not reach 50 new fabs by 2030, having 5 to 10 more would still be a significant achievement and a major milestone for the country's semiconductor ambitions. India's focus on the automotive sector is well-placed, given the strong presence of automotive companies and the growing demand for semiconductors in electric vehicles, autonomous driving, and smart vehicle technologies. I am very optimistic India will reach the target of \$110 billion by 2030, or even more than that.

Which are the new areas where semiconductor demand will be augmented?

Another crucial area of focus is healthcare, particularly smart healthcare. The role of semiconductors in medical devices and healthcare technology is to be considered. From wearable health monitors to advanced imaging equipment and personalised medical devices, semiconductors are the backbone of these innovations. The integration of AI, sensors, and connectivity in healthcare devices is revolutionising patient care, enabling real-time health monitoring, early disease detection, and more effective management of chronic conditions. India's commitment to focusing on these sectors will not only bolster its semiconductor industry but also contribute to improving healthcare systems domestically and globally. **EFY**